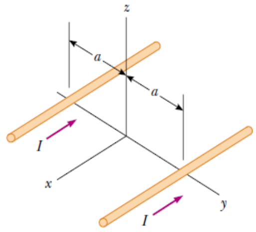
PH 102 (Jan-May-2024), Tutorial-8, Date: 12 March, 2024 (Tuesday 7:55-8:50 AM)

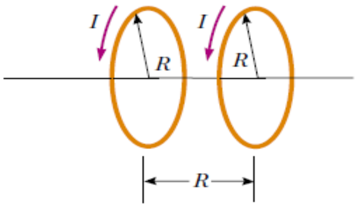
1. Two parallel conducting plates are a distance d apart and have a potential difference V between them. A uniform magnetic field B is also present in the region between the plates along a direction parallel to the plates. Particles of charge +q and mass m are released from rest in the crossed E and B fields from the surface of one plate. Find the trajectory of the particles.

2. A loop carrying current is shaped like a regular plane polygon of 2n sides, the distance between parallel sides being 2a. Find the magnetic field at the centre of the loop.

3. In Figure below, both currents in the infinitely long wires are in the negative *x* direction. (a) Sketch the magnetic field pattern in the *yz* plane. (b) At what distance *d* along the *z* axis is the magnetic field a maximum?



4. Two circular coils of radius *R*, each with *N* turns, are perpendicular to a common axis. The coil centers are a distance *R* apart. Each coil carries a steady current *I* in the same direction, as shown in Figure below.



(a) Show that the magnetic field on the axis at a distance *x* from the center of one coil is

$$B = \frac{N\mu_0 I R^2}{2}\left[\frac{1}{\left(R^2+x^2\right)^{3/2}} + \frac{1}{\left(2R^2+x^2-2Rx\right)^{3/2}}\right]$$

(b) Show that dB/dx and d^2B/dx^2 both are zero at the point midway between the coils. This means the magnetic field in the region midway between the coils is uniform.

5. The surface current density in the xoy plane is given to be $\vec{K}(x,y) = K_0\left(\frac{x^2}{a^2}\hat{i} + \frac{y^2}{b^2}\hat{j}\right)$.Evaluate $\vec{\nabla}.\vec{K}$ and hence calculate the time rate of change of charge contained in the first quadrant of a circle of unit radius centred on the origin. Show explicitly that for this quadrant, the continuity equation is satisfied in its integral form.

6. Consider an infinitely long transmission line consisting of two concentric cylinders having their axes along the z-axis. The cross-section of the transmission line is shown in figure below where z-axis is out of the page. The inner conductor has radius a and carries current I while the outer conductor has inner radius b and thickness d and carries return current –I. Determine $\vec{B}$ everywhere, assuming that current is uniformly distributed in both conductors.